Viviane Alves de Oliveira

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SUMMARY

AI Engineer with a strong software engineering background and 10+ years of experience with **Python** and hands-on experience building end-to-end AI systems across healthcare, machine learning, audio, **NLP**, and **Generative AI**. Experienced in developing and deploying 2D/3D deep learning models, designing scalable APIs and ML pipelines, and building production **LLM, RAG**, and agentic AI systems using **PyTorch, TensorFlow, LangGraph, LangChain, FastAPI**, and Google Cloud. Strong experience in AI backend architecture, model and LLM integration, vector search, observability, and production AI workflows. Interested in building reliable AI systems while continuing to deepen expertise in model development, MLOps/LLMOps, and agentic AI.

PROFESSIONAL EXPERIENCE

Senior AI Software Developer 03/2026 – 08/2026
OpenSports Montreal, Canada

- Designed and implemented production **AI agents** and reasoning pipelines using **Python** and LangGraph, including state management, tool-selection logic, reflection cycles, node navigation, fallback strategies, and error handling.
- Integrated Gemini models through the **Vertex AI SDK**, configuring inference parameters and optimizing prompts for structured LLM interactions within production AI workflows.
- Implemented **RAG** pipelines and semantic search, including embedding generation, **BigQuery Vector Search**, and in-memory embeddings to support context retrieval for LLM-powered applications.
- Developed the **Node.js** AI gateway with asynchronous request handling and Server-Sent Events (SSE), enabling real-time streaming of LLM responses to client applications and improving perceived response latency.
- Designed and implemented LLM/agent observability with **Langfuse**, providing centralized tracing of prompts, token consumption, latency, errors, and chained **LangGraph** executions to support debugging and production monitoring.

Artificial Intelligence Engineer 01/2025 – 01/2026
Freelancer Montreal, Canada

- **Lisnen (Toronto, Canada) – Audio Recognition | 08/2025 – 01/2026**
  - Designed and trained audio classification models using CNN-based architectures and pretrained audio embeddings (YAMNet, PANNs) with **TensorFlow/Keras**, targeting real-world sound recognition use cases.
  - Built audio preprocessing and augmentation pipelines using Librosa and **Audiomentations**, and optimized models for real-time edge inference through **TensorFlow Lite** (TFLite).
- **Revelo (USA) – ML Traceability Pipeline | 04/2025 – 09/2025**
  - Developed end-to-end **ML pipelines** covering data ingestion, preprocessing, feature engineering, model training, evaluation, and deployment for traceability and analytics use cases.
  - Containerized ML services and workflows using **Docker** and **Django**, integrating data processing and model inference components into reproducible deployment pipelines.
- **Angelina (USA) – NLP / Sentiment Analysis | 01/2025 – 02/2025**
  - Developed a sentiment analysis solution using **NLP** techniques and transformer-based embeddings, including BERT/GPT-based approaches, for unstructured text classification.
  - Prepared noisy and imbalanced text datasets and developed production-oriented inference workflows with documented results for non-technical stakeholders.

Machine Learning Engineer (Intern) 05/2025 – 07/2025
BrainInnov Montreal, Canada

- Trained 3D CNN models (DenseNet, ResNet, Attention Mechanism variants) with **PyTorch**, optimizing learning rates using Optuna and addressing gradient instability.
- Improved evaluation through F1 score, precision-recall metrics, and visual interpretability (Grad-CAM).
- Applied class-aware data augmentation (on-the-fly and pre-saved hybrid) to manage memory and balance datasets.
- Built a preprocessing pipeline for DICOM-based 3D CT scans using **SimpleITK** and **Monai** library, supporting volume normalization, augmentation, and class balancing.

Python Developer 03/2022 – 07/2023
Globant Brasilia, Brazil

- Designed and maintained ETL pipelines using **Python, Jupyter Notebooks**, and **Azure Data Factory**, automating data ingestion and transformation.
- Worked with **AWS Glue, Athena, and S3** to manage data lakes and deliver clean datasets for analytics teams.
- Mentored team members, reviewed code, and maintained documentation to ensure cross-team alignment on international projects.

Full Stack Developer (Python / Flask) 12/2021 – 03/2022
Radix Engineering and Software Brasilia, Brazil

- Built an inventory monitoring application from scratch using **Python (Flask), jQuery**, and **Databricks**, improving inventory visibility for retail clients.
- Processed large-scale data with **PySpark**, integrating business rules into data transformation pipelines.
- Ensured application scalability and performance through continuous integration and testing.

AI BackEnd Software Developer (Python / Django) 12/2020 – 12/2021
Portal Telemedicine Brasilia, Brazil

- Delivered AI diagnostic outputs to doctors by integrating containerized microservices using **Python, Django, Docker, and Kubernetes**, improving availability and system resilience.
- Built and deployed REST APIs to transfer AI-generated heatmaps to the frontend, using **Django REST Framework and JavaScript (Vue.js)**.
- Used **Google Cloud Platform (BigQuery, Data Fusion, Cloud Storage, Looker)** to create dashboards, monitor file uploads, and build automated pipelines.
- Collaborated with the AI team to expose model outputs, working closely with **C# and containerized AI modules**, ensuring smooth cross-service communication.
- Managed Git workflows and code reviews using **Git, Bitbucket**, and practiced Agile methodology for continuous delivery.

Full Stack Developer (Python / Django) 07/2013 – 02/2020
Softplan Brasilia, Brazil

- Developed a Django-based web application to manage court processes, using **Python, Django, PostgreSQL**, and deployed to **Heroku**, streamlining case assignments by court and prosecutor.
- Contributed to the project lifecycle from design to deployment, gaining strong backend and DevOps experience.
- Maintained legacy systems and transitioned to newer tools as the company scaled its digital infrastructure.

EDUCATION

AEC – Artificial Intelligence and Machine Learning 2023 – 2025
LaSalle College Montreal, Canada

Postgraduate Studies in Information Technology Governance 2010 – 2011
Faculdades Integradas Promove de Brasília - Credential evaluated by MIFI (Québec) Brasília, Brazil

DEC in Data Processing 2007 – 2010
Institut des Sciences Exactes - Credential evaluated by MIFI (Québec) Brasilia, Brazil

SKILLS & KNOWLEDGE

- **Programming:** Python, JavaScript, SQL
- **AI & Machine Learning:** Deep Learning, CNNs, Computer Vision, NLP, Audio Classification, PyTorch, TensorFlow, Keras, MONAI, Scikit-Learn
- **Generative AI:** LLMs, RAG, AI Agents, Agentic AI, Prompt Engineering, LangGraph, LangChain, Embeddings
- **AI Software Engineering:** FastAPI, Django, Flask, REST APIs, Microservices, Node.js, Express, SSE, Docker
- **Cloud & Data:** Google Cloud, Vertex AI, BigQuery, PostgreSQL, PGVector, MongoDB, AWS, Azure, ETL/Data Pipelines
- **AI Observability & LLMOps:** Langfuse, LLM Tracing, Prompt Monitoring, Model Evaluation, AI/ML Deployment, CI/CD
- **Testing & Development:** PyTest, Unit Testing, Git, GitHub, Bitbucket, Code Review, Agile/Scrum
- **Languages Skills**: Intermediate in progress: French Advanced: English Native: Portuguese